Random variables, concentration and the law of large numbers

Lecture and worked examples

Maths Explorer

Objectives

Compute the expected value and variance of a sum and of a mean; apply the Bienaymé-Chebyshev inequality; determine a sample size.

Sums and transformations

For a finite random variable X, taking the values x_i with probabilities p_i,

$$E(X) = \sum_i x_i p_i, \qquad V(X) = \sum_i (x_i - E(X))^2 p_i = E(X^2) - E(X)^2.$$

The standard deviation is $\sigma(X) = \sqrt{V(X)}$. When the square in $V(aX + b)$ is expanded, the shift b cancels and the factor a is squared. For finite random variables X, Y and real numbers a, b,

$$E(X + Y) = E(X) + E(Y), \quad E(aX + b) = aE(X) + b, \quad V(aX + b) = a^2 V(X).$$

Linearity of expected value does not require independence. If X, Y are independent, $V(X + Y) = V(X) + V(Y)$ (accepted without proof). Without this assumption, this formula cannot be applied. The standard deviation of $aX + b$ is $|a|\sigma(X)$.

Sums and transformations

If $Y = X$, then $V(X + Y) = V(2X) = 4V(X)$, and not $2V(X)$ as soon as $V(X) \neq 0$.

Sums and transformations

Write $X = B_1 + \cdots + B_n$ with the independent indicator variables of the successes. Each B_i satisfies $E(B_i) = p$, $E(B_i^2) = p$ and $V(B_i) = p - p^2$. By linearity and additivity under independence, $E(X) = np$, $V(X) = np(1-p)$.

Sample and mean

A sample of size n from a distribution is a family of n independent random variables with the same distribution. If they have expected value μ and variance σ^2, we write $S_n = \sum_{i=1}^{n} X_i$ and $M_n = \dfrac{S_n}{n}$.

Sample and mean

$$E(S_n) = n\mu, \quad V(S_n) = n\sigma^2, \qquad E(M_n) = \mu, \quad V(M_n) = \frac{\sigma^2}{n}, \quad \sigma(M_n) = \frac{\sigma}{\sqrt{n}}.$$

Sample and mean

To divide the standard deviation of the mean by 2, multiply the size by 4. The variance of the sum increases; that of the mean decreases.

Concentration

For a variable with expected value μ, variance σ^2, and $\delta > 0$,

$$P(|X - \mu| \geq \delta) \leq \frac{\sigma^2}{\delta^2}.$$

An upper bound greater than 1 remains true, but gives no additional information; it can be replaced by 1.

Concentration

In $V(X) = \sum_x (x-\mu)^2 P(X=x)$, keep only the values satisfying $|x-\mu| \geq \delta$. Each square is then at least δ^2. Hence $V(X) \geq \delta^2 P(|X-\mu| \geq \delta)$.

Law of large numbers

Applied to the mean of a sample,

$$P(|M_n - \mu| \geq \delta) \leq \frac{\sigma^2}{n\delta^2} \xrightarrow[n\to\infty]{} 0.$$

This is the law of large numbers in the setting considered: for a fixed precision, the probability of a deviation at least equal to that precision tends to zero. This result does not say that each realisation gets closer to μ monotonically.

Law of large numbers

To guarantee $P(|M_n - \mu| < \delta) \geq 1 - \alpha$, it suffices to choose

$$n \geq \frac{\sigma^2}{\alpha \delta^2}.$$

Take the ceiling. This is the smallest integer satisfying *this bound*, not necessarily the minimum size for the exact probability.

Law of large numbers

For a frequency of successes, $\mu = p$ and $\sigma^2 = p(1-p) \leq \frac{1}{4}$. If p is unknown, $n \geq \frac{1}{4\alpha\delta^2}$ is a uniform guarantee.

Research activity

Fluctuations of a frequency - 40 min.

A fair coin is tossed independently. For every integer $n \geq 1$, let F_n denote the frequency of "heads" among the first n tosses.

Research activity (2)

1. Write a program that simulates 1000 tosses, then plot the points $(n; F_n)$ for $1 \leq n \leq 1000$. Does the distance from $\frac{1}{2}$ necessarily decrease at each toss?
2. Express $E(F_n)$ and $V(F_n)$ in terms of n.
3. Using Bienaymé-Chebyshev, find an upper bound for $P(|F_{1000} - \frac{1}{2}| \geq 0.05)$.
4. Determine an integer n large enough that $P(|F_n - \frac{1}{2}| < 0.05) \geq 0.99$.

Research activity (3)

5. State what the simulated trajectory shows and what the inequality obtained guarantees. Does a single simulation constitute a proof of this guarantee?